

Admin & Super Admin Console

1. Feature Name

Admin & Super Admin Console

2. Purpose

- **Problem Solved:** Platform operators need central oversight: user administration (roles, statuses), platform statistics, learning-path visibility, certificate management, and classroom management.
 - **Why It Exists:** The `Admin` role gates a controller with full user/classroom/certificate CRUD, while a two-tier privilege model (`IsSuperAdmin` flag + roles) restricts the most sensitive operations (role changes) to the Super Admin only.
 - **Role in Application:** It is the governance/operations subsystem for the whole platform, layered on top of all other domains.
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3. What the User Can Do

Admin

- View platform stats: total users, learning paths, classrooms, certificates issued, modules completed, new users this month, and a 6-month user-growth series.
- List and search all users; view a single user's full profile + counters.
- Search/filter/paginate certificates (search by name/email/path title; filter by issue-date range).
- View a certificate detail and delete certificates.
- Search/filter/paginate classrooms (search by title/code/path/creator; filter by learning path and status).
- View classroom detail (members, roles, statuses) and edit classroom title/description.
- Reassign a classroom to a new learning path.
- Delete classrooms and users.
- Mark users invalid (requires a reason), activate/deactivate users.
- View all learning paths.
- (Super Admin only) Change a user's role (Admin / Instructor / Student).

Super Admin

- Everything an Admin can do, **plus** changing user roles. Role endpoints reject non-Super-Admin actors.

4. Feature Workflow

[Admin opens console] → GET /api/v1/admin/stats

→ aggregates platform-wide counts + 6-month growth series

[Admin manages **users**]

→ GET /admin/users?search= → role/status/counters per user

→ PUT /admin/users/{id}/role (Super Admin only)

→ PUT /admin/users/{id}/invalid {reason} (Super Admin or Instructor)

→ PUT /admin/users/{id}/activate | /deactivate

→ DELETE /admin/users/{id} (soft delete + revoke tokens)

[Admin manages certificates]

→ GET /admin/certificates?search&fromDate&toDate&page&pageSize

→ GET /admin/certificates/{id} → DELETE /admin/certificates/{id}

[Admin manages classrooms]

→ GET /admin/classrooms?search&learningPathId&status&page&pageSize

→ GET /admin/classrooms/{id} → PUT /admin/classrooms/{id}

→ PUT /admin/classrooms/{id}/learning-path

→ DELETE /admin/classrooms/{id}

5. How It Works Internally

Platform Stats (GetStatsAsync)

Counts users, learning paths, classrooms, certificates, and completed progresses (Progresses.IsCompleted). `NewUsersThisMonth` counts users created since the 1st of the current month. `UserGrowth` builds a 6-month series by looping `i` from 5 down to 0: for each month start (YY-MM), counts users with `CreatedAt` in `[start, start+1month)` and labels the entry with `start.ToString("MMM")`.

User Listing (GetAllUsersAsync)

- Optional search over email, username, first/last name (substring).
- Sorted `CreatedAt` desc.

- Last-login heuristic: `RefreshTokens` grouped by `UserId`, taking `Max(CreatedAt)` per user, joined into a dictionary — the latest refresh-token issuance approximates the last login.
- Per-user DTO (`BuildUserDtoAsync`) adds live counters: paths created, modules completed, quiz attempts, certificates, classrooms joined.

Role Change (`UpdateUserRoleAsync`)

- **Guard:** actor must have `IsSuperAdmin == true` (role field is otherwise ignored because the controller allows any `Authenticated+Admin` and permission is enforced inside the service).
- **Guard:** target user cannot be the Super Admin account (`user.IsSuperAdmin` → reject).
- Valid roles: `Admin`, `Instructor`, `Student` — anything else is rejected.
- Removes current roles, adds the new role, logs `ROLE_CHANGED` (old/new values) plus conditional `INSTRUCTOR_ASSIGNED` / `INSTRUCTOR_REMOVED` audits.

Mark Invalid / Activate / Deactivate (`MarkInvalidAsync` / `ActivateUserAsync` / `DeactivateUserAsync`)

- **MarkInvalid:** allowed for Super Admin or any `Instructor` role (checked via `IsInRoleAsync`). A non-empty `Reason` is required. Sets `Status = Invalid`, stores `InvalidReason`, logs `USER_MARKED_INVALID`.
- **Activate:** restores `Status = Active`, clears `InvalidReason`; logs distinct audit messages depending on whether the user was `Invalid` (`INVALID_REMOVED` vs `USER_ACTIVATED`).
- **Deactivate:** sets `Status = Inactive`, logs `USER_DEACTIVATED`. Both activate/deactivate refuse a user already in `Deleted` status.

Soft Delete (`DeleteUserAsync`)

- Refuses the Super Admin and already-deleted users.
- **Soft delete** (no hard delete): renames `UserName` → `deleted_user_{id}`, sets `Email` → `deleted_{id}@deleted.local`, blanks `FirstName/LastName/Bio/AvatarUrl`, clears `EmailConfirmed`, sets `Status = Deleted`.
- Revokes all non-revoked `RefreshTokens` for the user so any active sessions die.
- Logs `USER_SOFT_DELETED`. Identity constraints require the unique username/email to change before reuse — hence the renames.

Certificates (controller-level, direct EF + auditing)

- Paged/searchable listing (`page ≥ 1`, `pageSize` clamped 1–100). Search matches first name, last name, full name, email, or path title. Date filters (`fromDate` / `toDate`) parsed via `DateTime.TryParse` and applied to `IssuedAt`.
- Certificate number is derived, not stored: `$"CERT-{Id:D6}"` (zero-padded 6-digit). `CompletedAt` mirrors `IssuedAt`.
- Deletion removes the row and writes a `CERTIFICATE_DELETED` audit.

Classrooms (controller-level, direct EF + auditing)

- Paged/searchable listing; search matches title, invite code, path title, creator name/email; optional filter by `learningPathId; status == "Archived"` intentionally matches nothing (no archived state today — filter reserved for future).
- `MemberCount = c.UserClassrooms.Count`; every classroom is reported `Status = "Active"`.
- Detail loads members (ordered by join date) with role + user status + invalid reason.
- Update requires a title; can optionally reassign `LearningPathId` (validated to exist) and `TrainerId` (validated to exist).
- `ReassignLearningPath` refuses no-op reassignment, logs old → new path titles.
- Deletion removes the classroom and logs `CLASSROOM_DELETED` with path title. `JOINED/LOCKED` status of memory: classroom delete is cascading via EF relationships.

Super Admin Model

`User.IsSuperAdmin` is a **boolean flag** separate from Identity roles. `RoleSeeder` seeds the seed admin account with `IsSuperAdmin = true` and keeps it true on subsequent runs. Role changes and invalid marking for the super-admin account are blocked. The frontend distinguishes Super Admin vs. Admin, and the audit-log viewer is Super-Admin-gated (feature 14).

6. Frontend Implementation

Layer	File Path	Responsibility
Page	<code>frontend/src/pages/admin/AdminPage.tsx</code>	Layout/nav shell + stats dashboard
Page	<code>frontend/src/pages/admin/AdminUsersPage.tsx</code>	User list, search, role/status management UI
Page	<code>frontend/src/pages/admin/AdminCertificatesPage.tsx</code>	Certificate list/detail/delete
Page	<code>frontend/src/pages/admin/AdminClassroomsPage.tsx</code>	Classroom list/detail/edit/reassign/delete
Page	<code>frontend/src/pages/admin/AdminPathsPage.tsx</code>	All learning paths
Page	<code>frontend/src/pages/admin/AdminHistoryPage.tsx</code>	Audit log viewer (Super Admin gated)
Pages	<code>frontend/src/pages/admin/AdminQuizManagementPage.tsx</code> , <code>AdminQuizEditorPage.tsx</code> , <code>AdminQuizAnalyticsPage.tsx</code> ,	Quiz/path content administration

Layer	File Path	Responsibility
	AdminQuestionBanksPage.tsx , AdminModuleEditorPage.tsx	

7. Backend Implementation

Layer	File Path	Responsibility
Controller	backend/Controllers/AdminController.cs	api/v1/admin/* endpoints (stats, users, certificates, classrooms)
Service	backend/Services/Admin/AdminService.cs	Stats, user management, role transitions, soft delete, paths
Interface	backend/Interfaces/Services/IAdminService.cs	Service contract
DTO	backend/DTOs/Admin/AdminDto.cs	Admin response/request models
Seeder	backend/Data/Seeders/RoleSeeder.cs	Super Admin seed account + flag maintenance

8. Database Implementation

Entity/Table	Purpose
AspNetUsers (+ IsSuperAdmin, Status, InvalidReason)	User lifecycle state; Status enum Active/Inactive/Invalid/Deleted
RefreshTokens	Last-login heuristic + session revocation on soft delete
Certificates	Certificate listing, search, delete
Classrooms / UserClassrooms	Classroom listing, detail, members
LearningPaths / Progresses / QuizAttempts	Stats + per-user counters

Migrations of note: `20260731094731_AddUserStatusAndSuperAdmin.cs` (adds `Status`, renames/backfills `IsSuperAdmin`, seeds flag), `20260731101137_AddInvalidStatusAndQuizOwnership.cs` (adds invalid status + quiz ownership backfill to the Super Admin).

9. Security & Authorization

- Whole controller is `[Authorize(Roles = "Admin")]` — the endpoint surface is Admin-only.
 - **Role change** additionally requires the actor's `IsSuperAdmin` flag inside the service; the Super Admin account's own role is immutable.
 - **Mark invalid** requires Super Admin *or* the `Instructor` role.
 - **Soft delete / activate / deactivate** — Admin only, with Super-Admin-protection and deleted-user guards.
 - `KeyNotFoundException` → 404; `ArgumentException` → 400; `UnauthorizedAccessException` → (403 via middleware) for guarded actions.
 - All mutations emit audit log entries (see feature 14).
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10. Important Business Rules

1. Only the **Super Admin** can change roles; the Super Admin account itself cannot be reassigned/deleted/marked invalid.
 2. Roles are limited to `Admin`, `Instructor`, `Student`.
 3. **Soft delete** preserves row integrity (rename to `deleted_user_{id}` + `deleted_{id}@deleted.local`) and revokes active refresh tokens.
 4. `MarkInvalid` requires a reason; deleted users cannot be activated/deactivated/invalidated.
 5. Certificate number is computed (`CERT-{Id:D6}`), not stored.
 6. Classroom `Archived` status filter matches nothing today (reserved).
 7. Last login is approximated by the max refresh-token issuance time.
 8. Certificates list date range is inclusive (`IssuedAt >= from` and `IssuedAt <= to`).
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11. Example of Internal Execution

Step 1: Super Admin changes a user's role

- `PUT /api/v1/admin/users/a1b2c3/role` with `{ role: "Instructor" }`, acting user `superadmin`.
- Service checks `actor.IsSuperAdmin` → pass; target is not super admin → pass; "Instructor" is valid → roles swapped; `ROLE_CHANGED` (+ `INSTRUCTOR_ASSIGNED`) audit rows written; updated DTO returned.

Step 2: Admin marks a user invalid

- `PUT /api/v1/admin/users/u7/mark-invalid` with `{ reason: "Plagiarism" }`.
- Actor is Admin (not super admin, not instructor) → the service **rejects** — only Super Admin or Instructor may mark invalid. 403 returned.

Step 3: Admin soft-deletes a user

- `DELETE /api/v1/admin/users/abc123`.
- User renamed to `deleted_user_abc123`, email → `deleted_abc123@deleted.local`, status `Deleted`, all refresh tokens revoked, `USER_SOFT_DELETED` audit written.

Step 4: Stats dashboard

- `GET /api/v1/admin/stats` — 6-month growth: `[("Mar", 4), ("Apr", 9), ("May", 12), ("Jun", 20), ("Jul", 31), ("Aug", 18)]` plus current totals.

12. Files Involved

Frontend

- `frontend/src/pages/admin/AdminPage.tsx`
- `frontend/src/pages/admin/AdminUsersPage.tsx`
- `frontend/src/pages/admin/AdminCertificatesPage.tsx`
- `frontend/src/pages/admin/AdminClassroomsPage.tsx`
- `frontend/src/pages/admin/AdminPathsPage.tsx`
- `frontend/src/pages/admin/AdminHistoryPage.tsx`
- `frontend/src/pages/admin/AdminQuizManagementPage.tsx`, `AdminQuizEditorPage.tsx`, `AdminQuizAnalyticsPage.tsx`, `AdminQuestionBanksPage.tsx`, `AdminModuleEditorPage.tsx`

Backend

- `backend/Controllers/AdminController.cs`
- `backend/Services/Admin/AdminService.cs`
- `backend/Interfaces/Services/IAdminService.cs`
- `backend/DTOs/Admin/AdminDto.cs`
- `backend/Data/Seeders/RoleSeeder.cs`

Database

- `backend/Entities/User.cs` (`IsSuperAdmin`, `Status`)
- `backend/Migrations/20260731094731_AddUserStatusAndSuperAdmin.cs`
- `backend/Migrations/20260731101137_AddInvalidStatusAndQuizOwnership.cs`

13. Functionality

- Platform-wide stats with 6-month user-growth series
- User search/list/detail with live activity counters
- Role assignment (Super Admin only; Admin/Instructor/Student)
- Mark-invalid with required reason (Super Admin or Instructor)
- Activate / deactivate users
- Soft delete with token revocation and identity renaming
- Certificate search/filter/paginate, detail, delete
- Classroom search/filter/paginate, detail, edit, reassign path, delete
- Learning-path list (all paths)
- Super Admin flag model + guarded endpoints
- Audit logging on every admin mutation